

Pattern Matching

15-3-26

Aim – Write a python script to prompt the user to enter a password. Use regular expressions to validate the password based on the criteria: atleast 8 characters long, contains atleast one uppercase letter, one lowercase letter, one digit, and one special character

Theory – Password validation is a process used to ensure that user-created passwords meet specific security requirements. It typically checks for length, character variety, and complexity to prevent weak passwords. Implementing such validation improves system security by reducing vulnerability to attacks like brute force and unauthorized access, ensuring safer authentication mechanisms. The re library is used to achieve this to use regular expression matching.

A1.

```
import re
```

```
print("Enter a password:")
```

```
password = input()
```

```
# Validate password criteria
```

```
has_length = len(password) >= 8
```

```
has_uppercase = re.search(r'[A-Z]', password)
```

```
has_lowercase = re.search(r'[a-z]', password)
```

```
has_digit = re.search(r'\d', password)
```

```
has_special = re.search(r'[!@#$%^&*()_+~\[\]{};:\'",.<>?/\|`~]', password)
```

```
if has_length and has_uppercase and has_lowercase and has_digit and has_special:
```

```
    print("Password is valid!")
```

```
else:
```

```
    print("Password is invalid. It must contain:")
```

```
    if not has_length:
```

```
        print("- At least 8 characters")
```

```
    if not has_uppercase:
```

```
        print("- At least one uppercase letter")
```

```
    if not has_lowercase:
```

```
        print("- At least one lowercase letter")
```

```
    if not has_digit:
```

```
print("- At least one digit")
```

```
if not has_special:
```

```
    print("- At least one special character")
```

```
Enter a password:  
AAbb123#@  
Password is valid!
```

```
Enter a password:  
1243A  
Password is invalid. It must contain:  
- At least 8 characters  
- At least one lowercase letter  
- At least one special character
```

Conclusion

This program successfully demonstrates regular expression matching.